26900526 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCSTA301 - SAMPLING TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **10** questions

Each question carries **3** marks

1. What is respondent bias? [K] / [CO1]
2. What is meant by the sampling fraction? [U] / [CO2]
3. When is proportional allocation preferred over optimum allocation? [K] / [CO3]
4. What are the disadvantages of cluster sampling? [U] / [CO5]
5. What are sampling frame errors? [K] / [CO1]
6. Why is SRSWR considered theoretically simpler but practically less convenient? [K] / [CO2]
7. What is the stratum weight? What is the sum of the stratum weights? [K] / [CO3]
8. Define sampling interval in circular systematic sampling? [K] / [CO4]
9. What is sampling error? [K] / [CO1]

10. What is the probability that a specified unit of a population of size N is included in a sample of size n based on SRSWOR? [U] / [CO2]

11. Write down the unbiased estimator of the variance of the population total under stratified random sampling. [K] / [CO3]

12. Under what condition does the variance of the systematic sample mean equal to the variance of the simple random sample mean? [K] / [CO4]

[3x10 = 30]

Part B

Short essay type questions

Answer any **4** questions

Each question carries **7** marks

13. Explain the features of Probability Sampling. [U] / [CO1]

14. Explain the confidence interval for the Population proportion P in the case of attributes. [K] / [CO2]

15. A population of size 800 is divided into 3 strata , their standard deviations are given below: [K] / [CO3]

Stratum	I	II	III
Size	200	300	300
SD	16	8	12

A stratified random sample of size 120 is to be drawn from this population. Determine the sizes of the samples from the strata adopting Neyman's optimum allocation.

16. Suppose 'N' units of a population are labelled serially from 1 to N. Write down the possible systematic samples in each of the cases when N=11, n=4 and N= 20 , n= 4. [K] / [CO4]

17. Critically evaluate the Lottery Method and the Random Number Table Method. Under what circumstances would you recommend each method? [U] / [CO2]

18. If N=20, n=4, k=5 and $Y_i = 2i$, find $V\left(\bar{y}_{sy}\right)$ and $V\left(\bar{y}_{srs}\right)$. [U] / [CO4]

[7x4 = 28]

Part C

Essay type questions

Answer any **1** questions

Each question carries **12** marks

19. Derive the formula for determining the sample size n for estimating the population proportion. [K] / [CO2]

20. What is Simple Random Sampling with Replacement (SRSWR)? Explain its procedure and properties with a suitable example. [K] / [CO2]

[12x1 = 12]